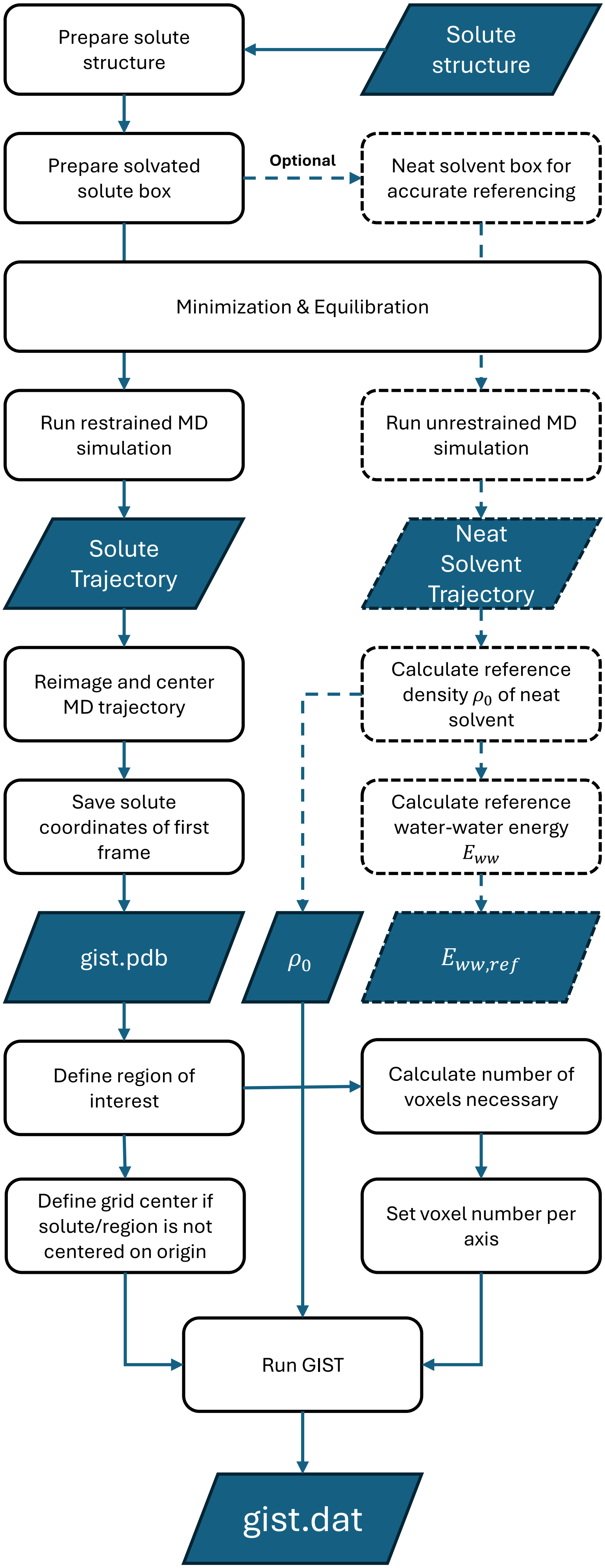


1. MD Simulation

❑ Add missing atoms

❑ Check protonation states

❑ Select force field

❑ Add solvent

At least 10-15Å buffer or 3 solvation layers depending on solvent

❑ Optional: create neat solvent box and determine NN entropy bias

❑ Restrain heavy atoms for solute simulation

Harmonic restraint potential $V_r(x) = k_r \cdot \Delta x^2$ with k_r =10 - 100 kcal mol^{-1} Å^{-2}

❑ Run simulation

≥10 ns (100 ns ideal)

≥ 10 000 frames (100 000 ideal)

❑ Confirm rigidity of solute heavy atoms: Suggested RMSD ≤ 0.1Å

2. Prepare GIST settings

❑ Determine reference water density ρ_0

Calculate reference from neat simulation or use tabulated value for the solvent model.

❑ Determine grid size

a) binding site

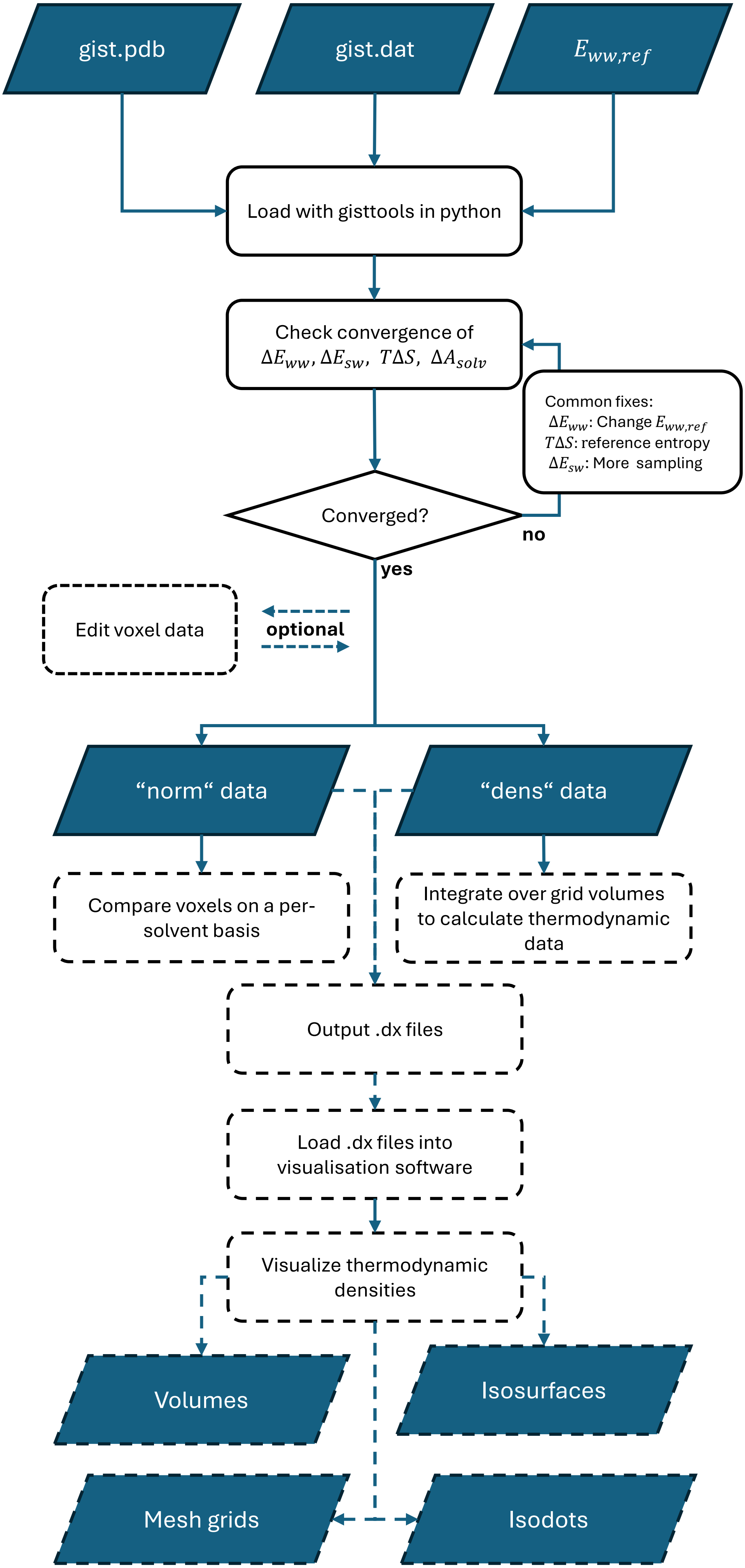
b) full system

c) 10 Å around region of interest

❑ Determine voxel number

For each cartesian direction, the number of voxels necessary is the length of the region of interest divided by the voxel side length (0.5 Å).

3. Run GIST in cpptraj



4a. Postprocess & Analysis

- ❑ Determine $E_{ww,ref}$
  Use tabulated value, calculate from reference simulation or autoreference from outer grid voxels, if the grid is large enough (> 3 solvent layers)
- ❑ Load gist.dat with gisttools
  Preset Eww,ref or use the autoreferencing. A pdb structure of the solute without water is necessary for this.
- ❑ Check convergence of thermodynamic properties – integrated values should approach a fixed value with distance from the region of interest (see Figure 4)
- ❑ Optional: edit voxel data
  For example, apply the empirical entropy correction factor of 0.6.
- ❑ Use “norm“ data to compare voxel to voxel
- ❑ Use “dens“ data to integrate over voxels or for visualisation purposes

4b. Visualisation (optional)

- ❑ If properties were modified or newly created, output density files for visualisation
- ❑ Visualize .dx files
  e.g. with PyMOL or VMD